

SEMESTER VII

UEI801 ADVANCED PROCESS CONTROL

L	T	P	Cr
3	1	2	4.5

Course Objectives: To make the students understand the basic concepts of advanced process control schemes, DCS, Artificial intelligence techniques used in Process Control, PLC and digital control system. To introduce artificial intelligence in process control. To explain programmable logic controller.

Introduction to advanced Control Schemes: Inferential control, Dead time and Inverse response compensation, Adaptive control, Model reference adaptive control, Self-tuning regulator Interactions and Decoupling of Control Loops: Design of cross controllers and selection of loops using Relative Gain Array.

Supervisory Control and Data Acquisition (SCADA) and Distributed Control System (DCS): Evolution and advantages of computer control, Configuration of Supervisory, Direct digital control (DDC), SCADA and DCS, Piping and instrumentation (PI) diagrams.

Artificial Intelligence in Process Control: Application of Expert systems, Neural networks, Fuzzy logic, Neuro-Fuzzy, Genetic algorithm in process control, tuning of PID controllers using intelligent methods.

Programmable Logic Controllers (PLC): Comparison with hard-wired relay and semiconductor logic, Hardware, Components of PLC, input-output modules of PLC, PLC memory design, PLC operation and communication, PLC troubleshooting and logic, switches, counters and timers, Ladder diagram programming, working logic of PLC by example, Case studies, Introduction to Simple Programmable Logic Device (SPLD), Complex Programmable Logic Device (CPLD), Field Programmable Gate Array (FPGA), logic cell, FPGA design and programming, netlist and Register Transfer Level (RTL), scan chains.

Instrumentation in Hazardous Areas: Introduction, hazardous area classification, protective concepts, enclosure classification designations, ingress protection rating, National Electrical Manufacturer Association (NEMA) types, equipment design and construction, intrinsically safe design, determining safe energy levels, intrinsic safety circuit design, installing intrinsically safe systems, transformer isolation barriers, relevant Indian standards.

Digital Process Control: Sampling and reconstruction in view of process control, Discrete systems analysis, Stability and controller design using z transform and difference equations, Smoothing filter realization in process control applications using difference equations, Ringing in digital controllers.

Course Learning Outcomes (CLO): After the completion of the course, the student will be able to:

1. Elucidate and apply the concept of advanced control schemes used in process control
2. Select, program and develop the application of distributed control system and PLC as in process control
3. Apply the knowledge of CPLD, SPLD, FPGA and artificial intelligence techniques in process control

4. Apply the knowledge of digital control system
5. Elucidate the concepts of instrumentation in hazardous area

Text Books:

1. *Stephanopoulos, G., Chemical Process Control, Prentice Hall of India Private Limited (2015).*
2. *Liptak, B.G., Instrument Engineers Handbook, Chilton Book Company (2012).*

Reference Books:

1. *Deb, S.R., Robotics Technology and Flexible Automation, Tata McGraw Hill (2017).*
2. *Johnson, C.D., Process Control Instrumentation Technology, Prentice Hall of India Private Limited (2005).*
3. *Zaidi, A., SPC Concepts, Methodologies and Tools, Prentice Hall of India Private Limited (1995).*

Evaluation Scheme:

S.No.	Evaluation Elements	Weightage (%)
1.	MST+EST	70
2.	Sessional (May include Assignments//Quizzes/Lab Evaluations)	30